La Scuola d'Italia / Huson / IB Mathematics Name:
12 February 2026

4.13 Classwork: Using trees to organize situations

1. Heinrik rolls two 6-sided dice at the same time. One die has three red sides and three black sides. The other die has the sides numbered from 1 to 6. By means of a tree diagram, table of outcomes or otherwise, answer each of the following questions.

(a) How many different possible combinations can he roll?

(b) What is the probability that he will roll a red and an even number?

(c) What is the probability that he will roll a red or black and a 5?

(d) What is the probability that he will roll a number less than 3?

Working:

Answers:
(a) …………………………………………..
(b) ....
(c) ....
(d) ……………………………………..........

(Total 8 marks)

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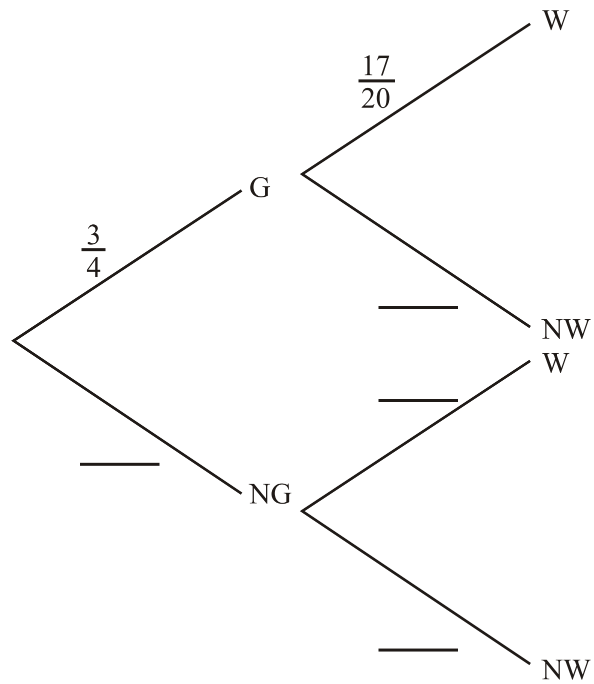
2. Today Philip intends to go walking. The probability of good weather (G) is $\frac{3}{4}$. If the weather is good, the probability he will go walking (W) is $\frac{17}{20}$. If the weather forecast is not good (NG) the probability he will go walking is $\frac{1}{5}$.

(a) Complete the probability tree diagram to illustrate this information.



(b) What is the probability that Philip will go walking?

Working:

Answer:

(b) ....

(Total 8 marks)

3. There are two biscuit tins on a shelf. The **red** tin contains three chocolate biscuits and seven plain biscuits. The **blue** tin contains one chocolate biscuit and nine plain biscuits.

(a) A child reaches into the **red** tin and randomly selects a biscuit. The child returns that biscuit to the tin, shakes the tin, and then selects another biscuit.

Find the probability that

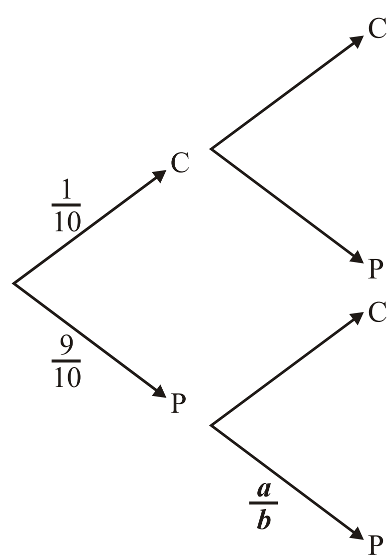
(i) both biscuits chosen are chocolate.

(2)

(ii) one of the biscuits is plain and the other biscuit is chocolate.

(3)

(b) A second child chooses a biscuit from the **blue** tin. The child eats the biscuit and chooses another one from the **blue** tin. The tree diagram below represents the possible outcomes for this event.



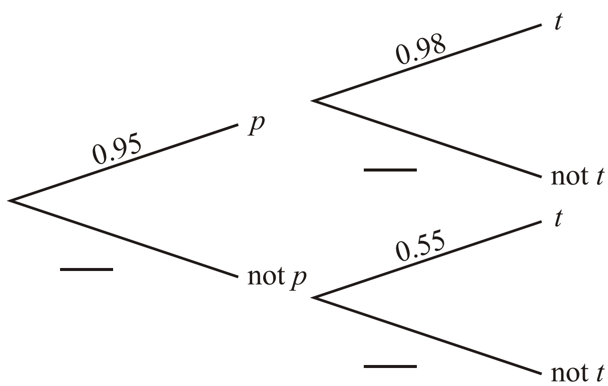
(i) Write down the values of ***a*** and ***b***.

4. The probability, p, that James gets up before 07.00 is 0.95.

If James gets up before 07.00, the probability, t, that he arrives at school on time is 0.98.

If James gets up later than 07.00, the probability that he arrives at school on time is 0.55.

The above information is represented by the following tree diagram.



(a) Complete the tree diagram.

(b) Calculate the probability that James gets up before 07.00 and is on time for school.

(c) Calculate the probability that James does **not** arrive at school on time.

Working:

Answers:

(b) ....

(c) ....

(Total 8 marks)